

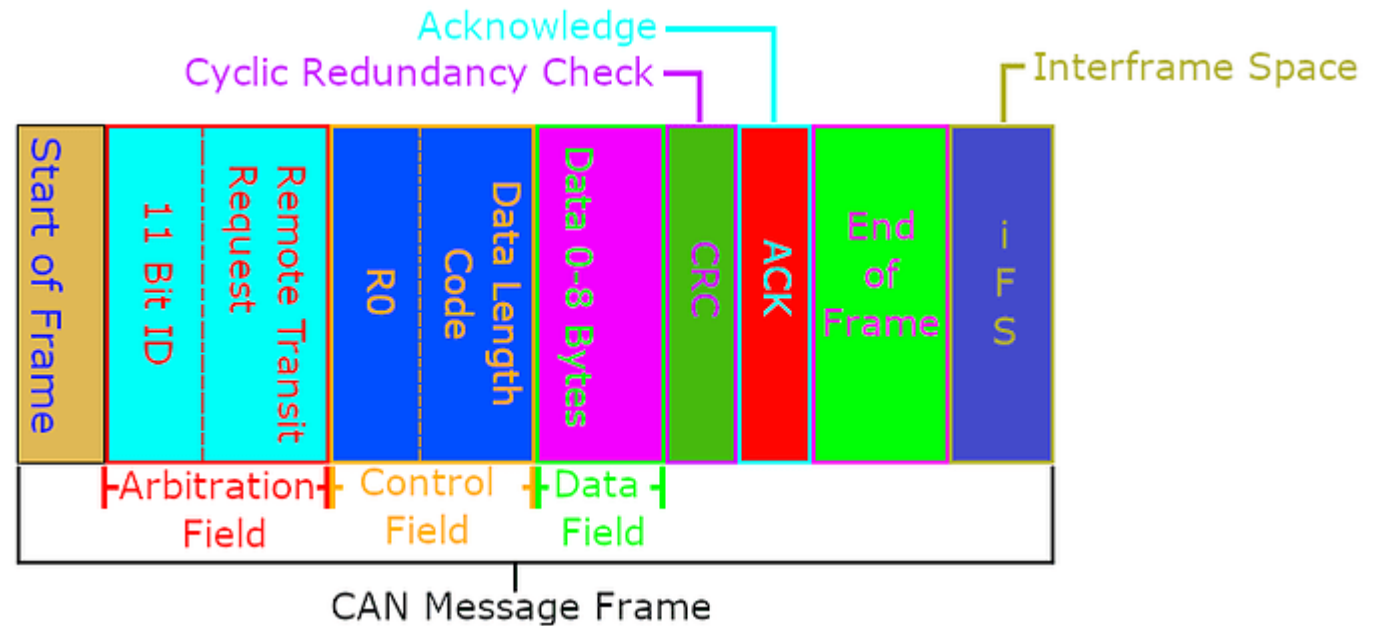
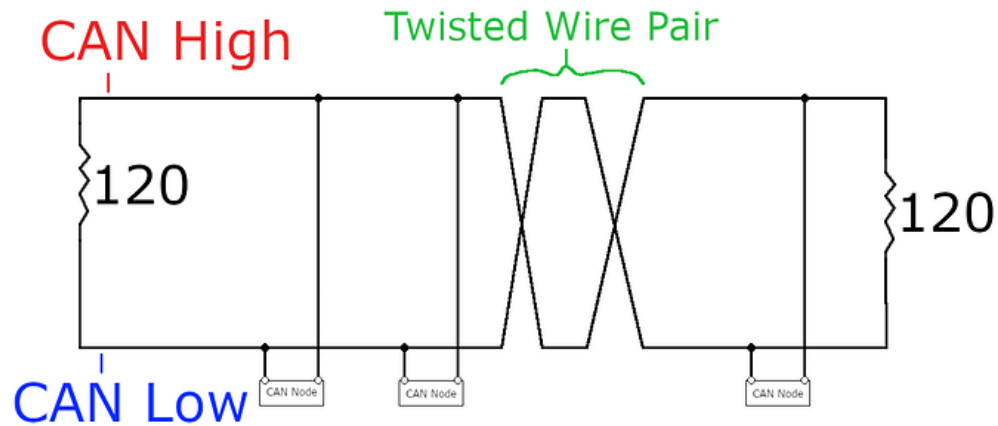
STM32 IoT Node Onboard Sensors

Main goals

- Read the BSP (Board Support Package) sensors and communicate with a host computer to visualize the data
- Background knowledge: serial communication protocols
 - **I2C**: Inter-IC Communication - **synchronous** serial protocol
 - **SPI**: Serial Peripheral Interconnect - **synchronous** serial protocol
 - **UART**: Universal **Asynchronous** Receiver and Transmitter
- BSP or off-board Sensors often use I2C, SPI, or UART to communication the main MCU.
- Other important protocol: **CAN** (Control Area Network) Bus
 - Also a kind of **Asynchronous** Serial Communication Protocol
 - Widely used in Vehicle Electronics
- Python Programming in a host computer (Linux/Windows/Mac)

CAN Bus

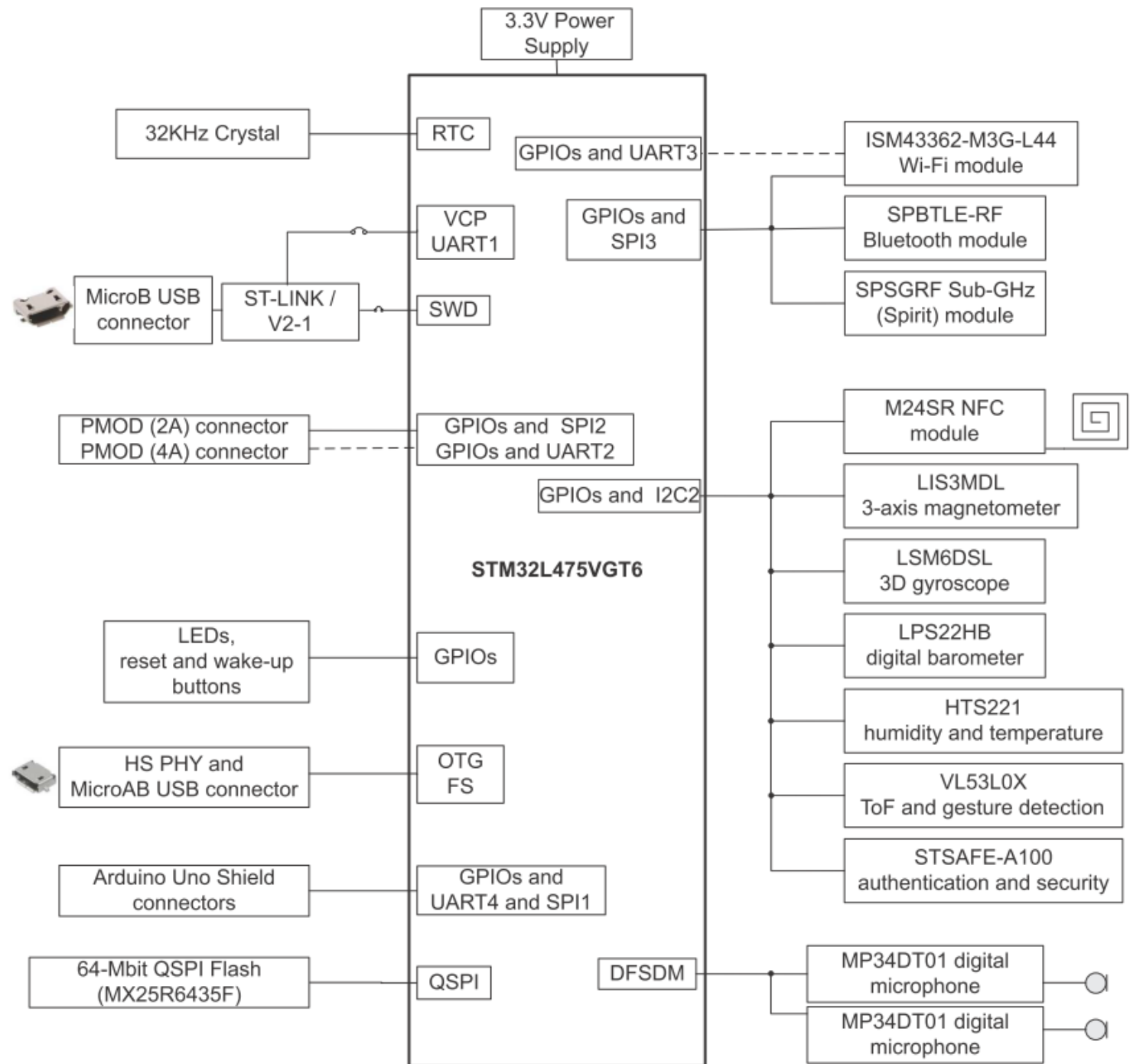
- CAN Bus protocol started with CAN 2.0, updated with ISO 11898, and developments are still being made (there are extended versions of the protocol for longer messages).
- An entire CAN Frame is typically 14 bytes long. Here is a breakdown of a CAN Message:
- CAN Bus use differential signals



B-L475E-IOT01A

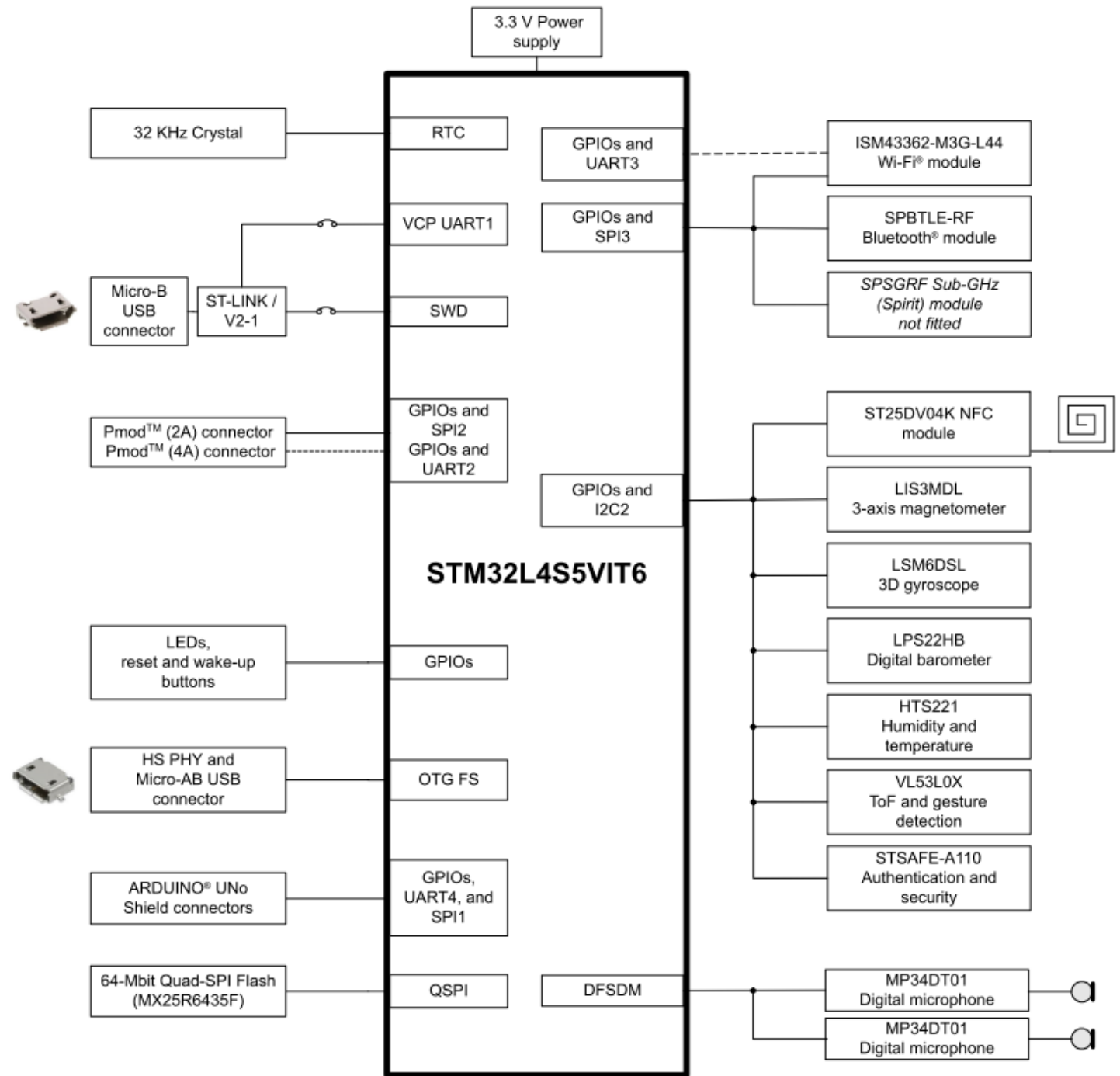
On board sensors

- Read on board sensors (temperature, humidity, pressure, gyroscope, accelerometer, magnetometer) using ST BSP drivers.



B-L4S5I-IOT01A on board sensors

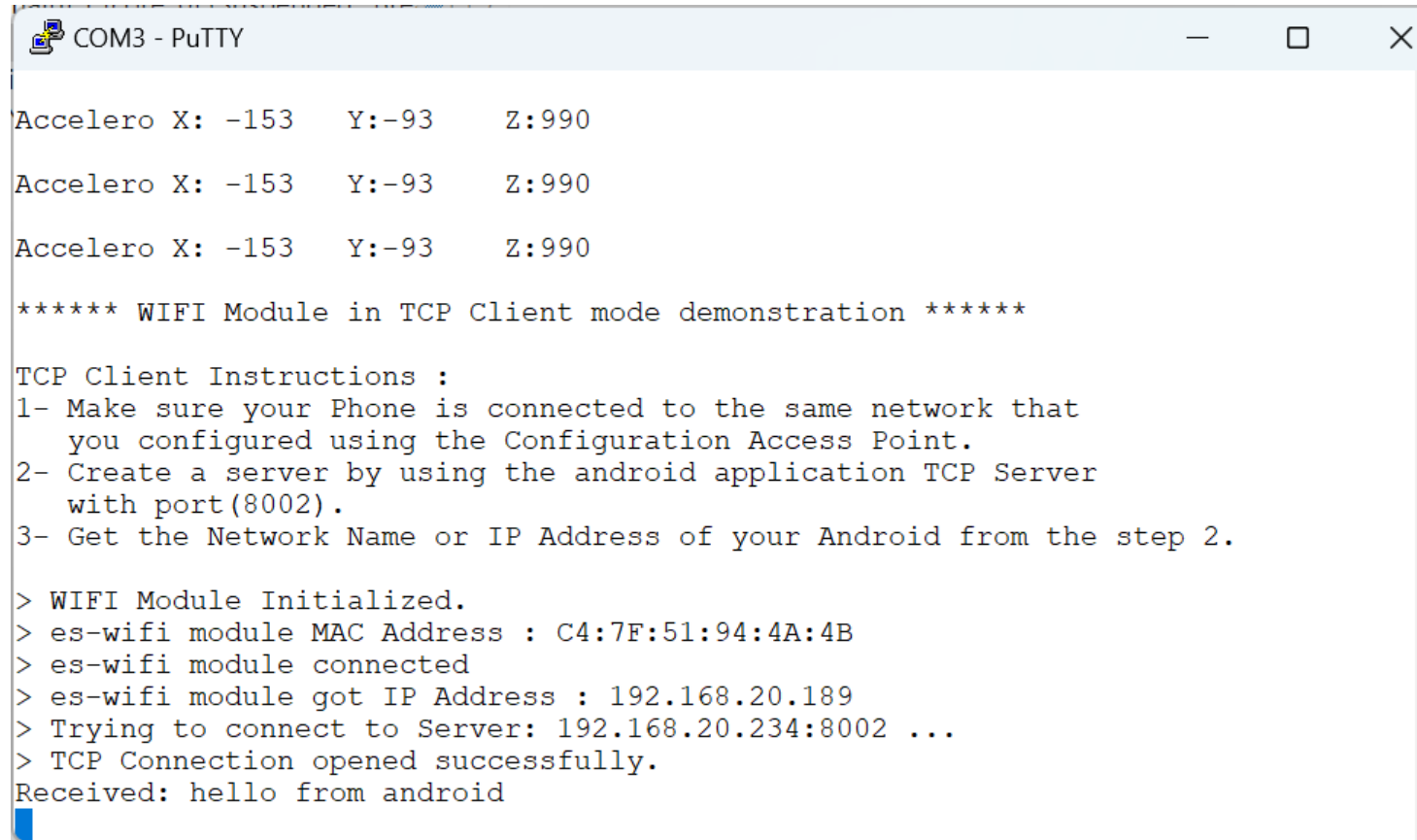
- 2020 version



Accelerometer readings

```
/* Includes -----*/  
#include "wifi.h"  
#include "stm32l475e_iot01.h"  
#include "stm32l475e_iot01_accelero.h"  
#include "stdio.h"
```

```
BSP_ACCELERO_Init();  
for (i=0; i < 20; i++)  
{  
    BSP_ACCELERO_AccGetXYZ(pDataXYZ);  
    TERMOUT("Accelero X: %d Y:%d Z:%d\n\n", pDataXYZ[0], pDataXYZ[1], pDataXYZ[2]);  
}
```



```
COM3 - PuTTY  
  
Accelero X: -153   Y:-93   Z:990  
Accelero X: -153   Y:-93   Z:990  
Accelero X: -153   Y:-93   Z:990  
  
***** WIFI Module in TCP Client mode demonstration *****  
  
TCP Client Instructions :  
1- Make sure your Phone is connected to the same network that  
   you configured using the Configuration Access Point.  
2- Create a server by using the android application TCP Server  
   with port(8002).  
3- Get the Network Name or IP Address of your Android from the step 2.  
  
> WIFI Module Initialized.  
> es-wifi module MAC Address : C4:7F:51:94:4A:4B  
> es-wifi module connected  
> es-wifi module got IP Address : 192.168.20.189  
> Trying to connect to Server: 192.168.20.234:8002 ...  
> TCP Connection opened successfully.  
Received: hello from android
```

References

- STM32 IoT node user manual
 - [Discovery kit for IoT node, multi-channel communication with STM32L4+ Series - User manual](#)
 - [B-L475E-IOT01A1](#) (2017-2019)
 - B-L4S5I-IOT01A (2020)
- Example Codes
 - https://github.com/iot2tangle/STM32_B-L475E-IOT01A
 - https://wiki.st.com/stm32mcu/wiki/STM32StepByStep:Step4_Sensors_usage